

SIO Math Workshop: Introduction to Statistics

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1 Introduction

• Why Statistics in Oceanography?

We might expect that the ocean should be deterministic. If we could properly measure the full state variables of every single water parcel at a given point in time, we should know exactly what the future state will look like. Unfortunately, we don't have anything remotely close to proper measurements of every water parcel.

- The ocean is complex. It's not possible to observe, measure, or specify every variable.
 - * Assuming there are 1.4×10^{24} g of sea water in the ocean, there are 4.7×10^{46} molecules to model.
 - * Initial conditions: three positions and three velocities for each molecule. Then we would need to specify 2.8×10^{47} numbers.
 - * We can't even do this on a computer. Assuming a laptop with 16GB RAM and single precision numbers, we would need 7.0×10^{37} laptops!
- Ocean observations are not controlled. All variables change as the system evolves.
 - * It's often impossible to design experiments that isolate a single variable.
 - * How do you decide what is "signal" and what is "noise"?
 - * There are many different temporal and spatial scales, and it's important to use techniques that are appropriate for the scales you're studying.
- Our measurements have only so much accuracy. They suffer from drift, noise, and resolution limitations. For nonlinear processes, the inaccuracies in our initial measurements can potentially grow into huge uncertainties in the future state.

Our incomplete view of the "initial state" of the ocean (at some time t_0) makes it impossible to unequivocally know what the future state will be. Therefore, we model the ocean as if it was a **random process** using statistics.

• Applications of Statistics

- Signal Processing
- Data Assimilation/Modeling
- Hypothesis Testing, identifying significant relationships
- Instrument Error

• Relevant SIO Classes

- SIOC 221A-B: Analysis of Physical Oceanographic Data
- SIOB 272: Advanced Statistical Techniques

- SIOG 223A-B: Geophysical Data Analysis
- SIOC 207A-D: Digital Signal Processing (Acoustics)
- SIOB 298: Bayesian Statistics
- ...and many others ...

2 Basic Probability and Set Theory

• Definitions

- Outcome, the specific result of an experiment
- Sample Space, Ω : the set of all possible outcomes of an experiment
- Event, E : a set of outcomes specified by some condition
- Probability, P : the extent to which an event is likely to occur; the ratio of “successes” to the total number of outcomes
 - * $P(E)$: probability of event E occurring

Outcome vs Events: An outcome is each of the individual results that an experiment could take. For example if you flip a coin three times

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

and an individual outcome could be

$$HTH.$$

An event is a set of outcomes specified by some condition. For example an event could be

$$E = \{\text{All outcomes with 2 heads}\} = \{HHT, HTH, THH\},$$

which is a set of three outcomes. Conversely if

$$E = \{\text{All outcomes with three tails}\} = \{TTT\},$$

this event corresponds to a single outcome.

• Probability Rules

- $P(\Omega) = 1$
- $0 \leq P(E) \leq 1$
- $P(\text{not } A) = 1 - P(A)$
- $P(A) \leq P(B)$ if A is a subset of B
- $P(A \text{ and } B) = P(A)P(B)$ if A and B are independent events
- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- Disjoint/mutually exclusive events
 - * $P(A \text{ and } B) = 0$
 - * $P(A \text{ or } B) = P(A) + P(B)$

• Class Example

- Rolling one fair die
 - * $\Omega = \{1, 2, 3, 4, 5, 6\}$, 6 total possible outcomes
 - * $P(6) = \frac{1}{6}$
 - * $P(< 3) = P(2 \text{ or } 1) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

• Practice Problems

- Rolling one fair die
 - * $P(\text{even})$
 - * $P(\text{not } 5)$
 - * $P(\text{not } < 3)$
- Rolling two fair dice
 - * Ω
 - * $P([2, 4])$
 - * $P(\text{both odd})$
- Flipping a fair coin three times
 - * $P(THH)$
 - * $P(\text{at least two heads})$
 - * $P(\text{exactly two tails})$

We can also define probabilities for multiple events.

• Definitions

- **Unconditional probability:** $P(A \text{ and } B)$, the probability that A and B are both true.
- **Conditional probability:** $P(A|B)$: the probability of A given that B is true

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} \quad (1)$$

- Generalized multiplication rule

$$P(A \text{ and } B) = P(A|B)P(B) = P(B|A)P(A) \quad (2)$$

- Statistical independence: when two variables have nothing to do with each other, then ...
 - * $P(A|B) = P(A)$
 - * $P(A \text{ and } B) = P(A)P(B)$
- Law of total probability:

$$P(A) = P(A|B)P(B) + P(A|\text{not } B)P(\text{not } B)$$

- Bayes' Theorem: relates the inverse of conditional probabilities

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (3)$$

• Class Example

- We know that 10% of days in the year have harmful algal blooms (HABs). Satellite monitoring for HABs has an accuracy of 80%, and a false positive rate of 15%. The monitoring system reports a HAB today; what is the probability there is actually a HAB?

We of course could have conditional probabilities defined in terms of three or more variables.

- $P(A, B|C)$: probability of A and B given that C is true
- $P(A|B, C)$: probability of A given that B and C are true

These conditional probabilities are related to the unconditional probabilities through similar rules as the 2-variable case. Specifically:

$$P(A, B|C) = \frac{P(A, B, C)}{P(C)}$$

and

$$P(A|B, C) = \frac{P(A, B, C)}{P(B, C)}$$

One obvious way of bringing more events into the mix is examining the probability of an event at some time t_N given that it occurred at prior times $\{t_1, t_2, \dots, t_{N-1}\}$.

$$P(A_{t_N} | A_{t_{N-1}}, A_{t_{N-2}}, \dots, A_{t_1})$$

3 Random Variables

– Definitions

- * Random variable, X : a function that maps all outcomes of an experiment to real numbers.

$$X : \Omega \rightarrow \mathbb{R}$$

Because we do not know which outcome in Ω will occur, the value of X is unknown until it is sampled. For time-dependent processes, the associated random variables have future states that cannot be determined by any collection of initial state information that we may have.

– Random variables vs. Events

- * Events are specific (possible) experimental results (e.g., “we will have a HAB tomorrow”, “flipping tails three times in a row”, “surface water temperature exceeds 14C at these two locations”)
- * Random variables allow us to quantify those results in measurable, real numbers (e.g., “the number of days we have HABs”, “the number of times we flip tails”, “the water temperature at the two locations”)
- Random variables allow us to better describe experimental results. They have properties described by probability rules, and open up many other types of analysis.
- Up until now, the probability we have been discussing is a map from events to numbers. If we want to extend our probability framework to random variables, we need a **probability distribution function**.

4 Probability Distributions

A probability distribution function p measures the likelihood of observing the possible values of a random variable. Note that since X maps Ω to real numbers, X has its own set of possible values called its state space or its range.

The same probability rules as before apply to the random variables as well. For example, if U is the random variable for the x component of velocity at some location and time, and the state space of U is $\mathbb{R}$, we would say $P(U \in \mathbb{R}) = 1$. However, the way in which probabilities are assigned to individual outcomes can differ slightly.

If X can only take discrete values, then its probability distribution function is a **probability mass function**. For an individual value $x \in X$, $p_X(x) = P(X = x)$ is the probability of observing x .

If instead, X can take on a continuum of values, its probability distribution function is a **probability density function**. For continuous random variables, the probability of observing any individual $x \in X$ is 0. The probability density function $p_X(x)$ is not equal to $P(X = x)$, rather it gives the relative likelihood of observing a value within an infinitesimal neighborhood of x .

- **Why do we need the probability distribution?**

The probability distribution is our segue into all sorts of statistical tests, methods of analysis, and comparison. From the probability distribution we can define **moments**, **cumulants**, **correlations**, **confidence intervals**, and so much more. In short, from knowing the probability distribution, we can start building the story as to why the system played out the way it did.

5 Empirical and Theoretical Probabilities

In order to gain access to all these fun statistical tools, we need the probability distribution for our random variables. How do we get it? There are two main ways:

- **Empirical probabilities:** The probability distribution is estimated from observations. If an experiment is repeated many times, the relative frequency of each outcome provides an estimate of its probability. This describes how the system *did* behave in the observations.
 - * Example: we roll a 6-sided die many times and make a histogram of outcomes. Given enough trials, the histogram should converge to a uniform distribution with each side occurring 1/6th of the time.
- **Theoretical probabilities:** The probability distribution is derived from a model, symmetry argument, or other physical/mathematical assumptions. This describes how the system *should* behave according to the theory.
 - * Example: the die has 6 sides, and assuming weight is evenly distributed within the die, there is no reason any side should be favored. Six sides with equal likelihood gives a uniform distribution with $p_X(x) = 1/6$ for each side.

Theoretical probabilities are great as both a shortcut and comparison tool. If the theory says that a certain phenomenon should have a probability distribution that agrees with our empirical probabilities, it is a good sign that we observed that phenomenon. However, theoretical probabilities can suffer from simplifications/assumptions in their derivations (e.g. nonlinear

phenomena likely won't match the linearized statistics) and may be unknown for certain phenomena.

Empirical probabilities on the other hand can always be constructed, and they do not require the sort of assumptions or simplifications of theoretical probabilities, but that does not guarantee they reflect the true probabilities of the underlying system (especially if those probabilities are time dependent). They generally are a mixture of multiple different processes (e.g. waves and turbulence), signal and noise, good data and bad data, etc. Therefore, it is helpful to validate with theoretical probabilities to help determine what all is dictating the probabilities in your dataset.

One example of theoretical probabilities that everyone should know is the **central limit theorem**. It should be covered in a later lecture, so I won't go into the details of where it comes from now. The point to be made is that through the CLT, the theoretical probabilities of the sum or mean of many random variables (subject to certain conditions) should be normally distributed.

As an example, the sea surface elevation η is a random variable given by the sum of amplitudes of many random surface waves. Those amplitudes themselves are random variables, and so under certain conditions, we expect η to be normally distributed.

Lastly, the theoretical probability of one random variable can be used to deduce the theoretical probability of other random variables through their mathematical relation.

- Example: U, V, W are independent and normally distributed components of velocity. The speed $\sqrt{U^2 + V^2 + W^2}$ will be chi distributed with 3 degrees of freedom.